

Representational Geometry in Neural Networks

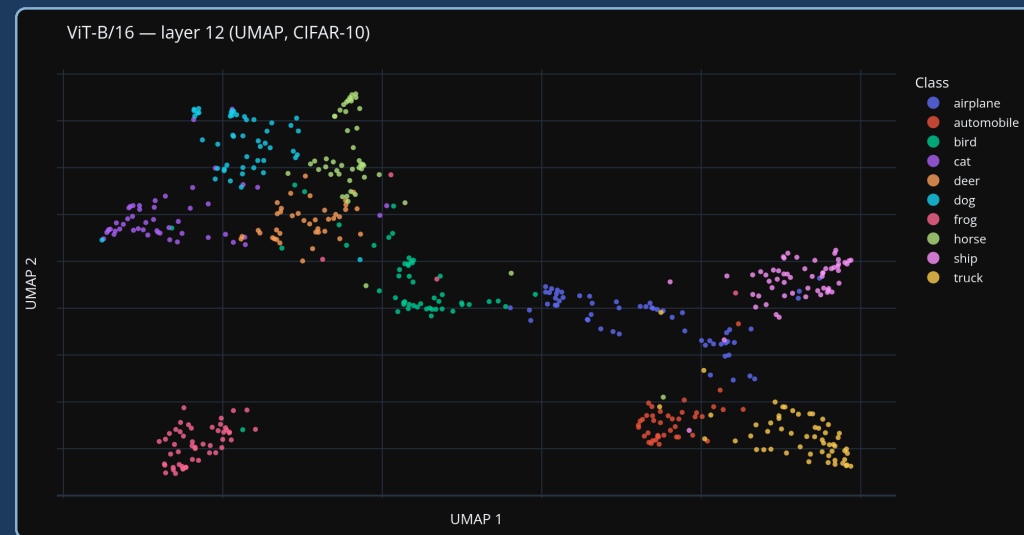
From Vision Transformers to RLHF-Aligned Language Models

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Summer Semester 2026

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ViT-B/16 · layer 12 · CIFAR-10 · real data

Part 1 — Cross-modal alignment

When a vision model and a language model train on the **same task**, does a shared representational structure emerge across modalities?

- **Hypothesis A:** cross-modal signal is new information → improves model
- **Hypothesis B:** models converge regardless — extra signal adds little

CKA before/after cross-modal fine-tuning · DINOv2 + Llama
· MS-COCO

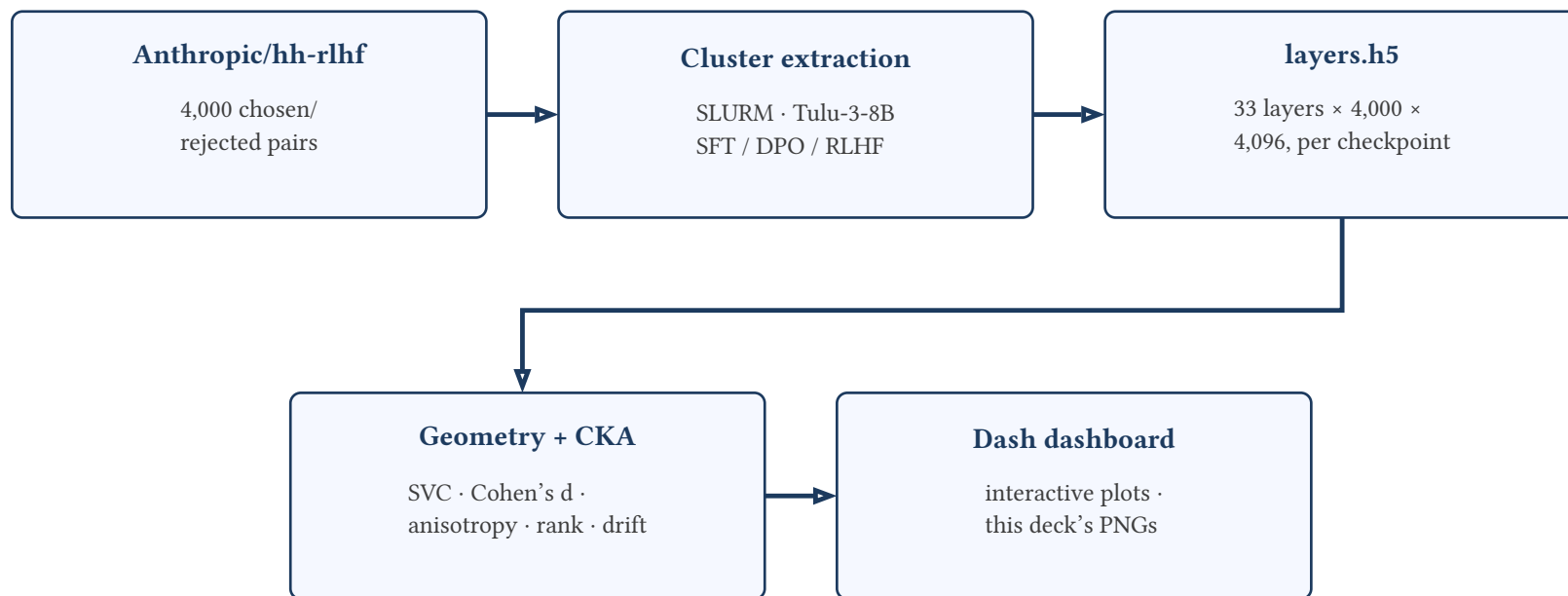
Part 2 — RLHF geometry

Does RLHF geometrically transform a language model's embedding space — encoding human preference as a **linearly separable structure**?

- Does alignment progressively separate chosen/rejected responses across SFT → DPO → RLHF?
- How does separation evolve **layer by layer**?

UMAP + LinearSVC per layer · Tulu-3-8B SFT/DPO/RLHF · Anthropic/hh-rlhf

Part 2 — Pipeline: From Dataset to Dashboard



`scripts/extract_embeddings.py` → `app/rlhf_geometry_compute.py` /
`app/vision_compute.py` → `app/app.py`

Part 2 — What Each Checkpoint Actually Is

Tulu-3-8B alignment trajectory — 3 public AllenAI checkpoints, same base model, extracted at each post-training stage

Definition: SFT Llama-3.1-

Tulu-3-8B-SFT

Supervised fine-tuning on curated instruction/response demonstrations. Model learns to follow instructions — no preference signal yet.

Definition: DPO Llama-3.1-

Tulu-3-8B-DPO

Direct Preference Optimization on chosen/rejected pairs. Directly increases $p(\text{chosen})$ over $p(\text{rejected})$ — no explicit reward model.

Definition: RLHF Llama-3.1-

Tulu-3-8B

Final RL stage on top of DPO (Tulu-3's RLVR — verifiable rewards on math/precise tasks). Labeled **RLHF** here for consistency with Part 1.

Remark All 3 are the **same** underlying 8B model at 3 successive training stages — not 3 different architectures. This is what makes the SFT → DPO → RLHF comparison a clean trajectory rather than a cross-model comparison.

Definition: Linear CKA For column-centered X, Y :

$$\text{CKA}(X, Y) = \frac{\|X^\top Y\|_F^2}{\|X^\top X\|_F \cdot \|Y^\top Y\|_F}$$

1 = proportional representations (no drift), 0 = orthogonal.

Definition: Cohen's d Direction $v = \frac{\mu_{\text{chosen}} - \mu_{\text{rejected}}}{\|\mu_{\text{chosen}} - \mu_{\text{rejected}}\|}$:

$$d = \frac{(\mu_{\text{chosen}} - \mu_{\text{rejected}}) \cdot v}{\sigma_{\text{pooled}}}$$

Effect size along the best linear separating direction.

Definition: Anisotropy Over randomly sampled pairs $i \neq j$:

$$\text{aniso}(X) = \mathbb{E}_{i \neq j} [\cos(x_i, x_j)]$$

Near 1 = representations collapse to a narrow cone.

Definition: Effective rank Participation ratio of covariance eigenvalues λ_i :

$$\text{erank}(X) = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}$$

How many dimensions are effectively in use.

Tulu-3-8B SFT → DPO → RLHF · Anthropic/hh-rlhf · 4,000 samples · 33 layers

Negative finding

Chosen/rejected preference is not linearly encoded in last-token pooled representations, at any alignment stage

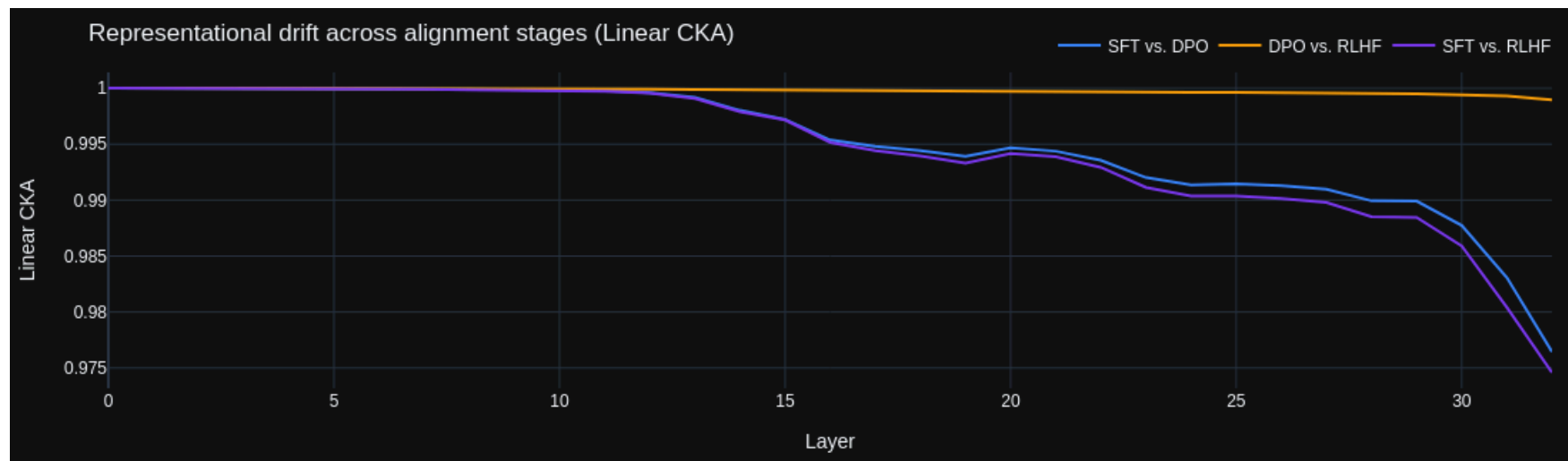
- Peak LinearSVC accuracy near chance in all 3 checkpoints — SFT 0.543, DPO 0.544, RLHF 0.550 (layer 14)
- Peak Cohen's d small and flat — 0.183 → 0.193 → 0.194
- No phase transition: alignment does not linearly reorganize preference geometry

But the geometry does shift

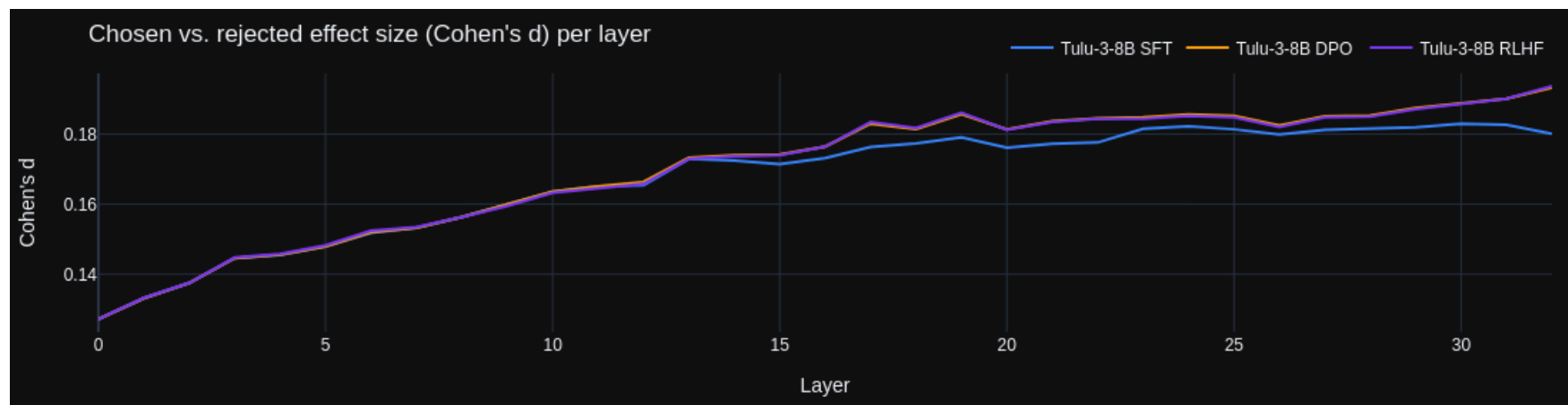
Representational drift (Linear CKA, same inputs, layer by layer):

- CKA(SFT, DPO) drops to **0.976** by layer 32 — most of the shift happens here
- CKA(DPO, RLHF) stays above **0.999** through layer 32 — RLHF/PPO barely moves geometry further
- Shift is concentrated at SFT→DPO — not a chosen/rejected separator, a general representational shift

Full curves (SVC, Cohen's d, CKA drift, anisotropy, effective rank) in the dashboard · Part 2 tab

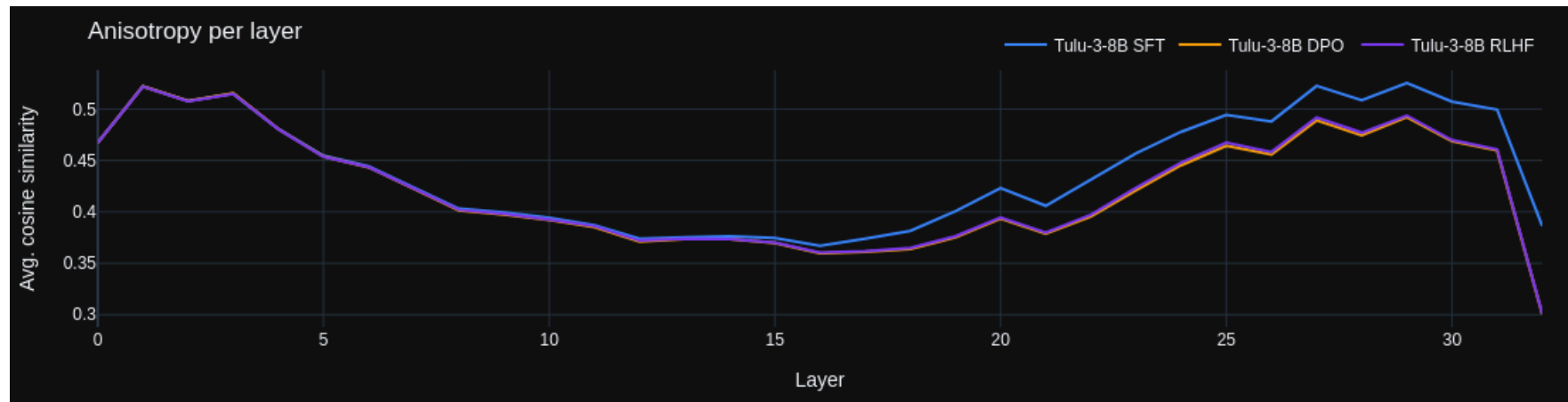


CKA drift — shift concentrated SFT to DPO, later layers

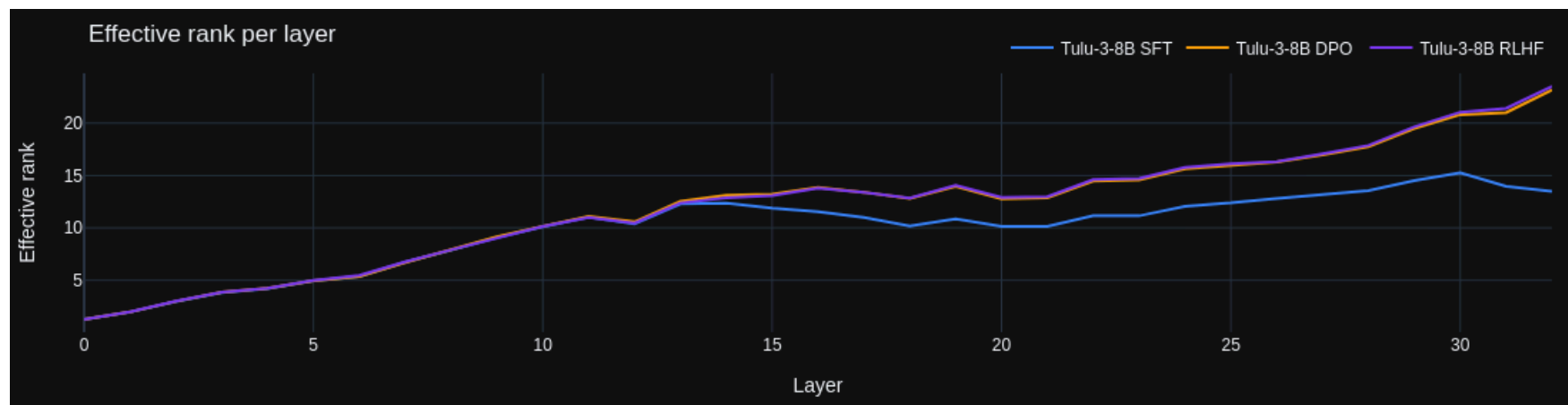


Cohen's d — small, flat across all layers

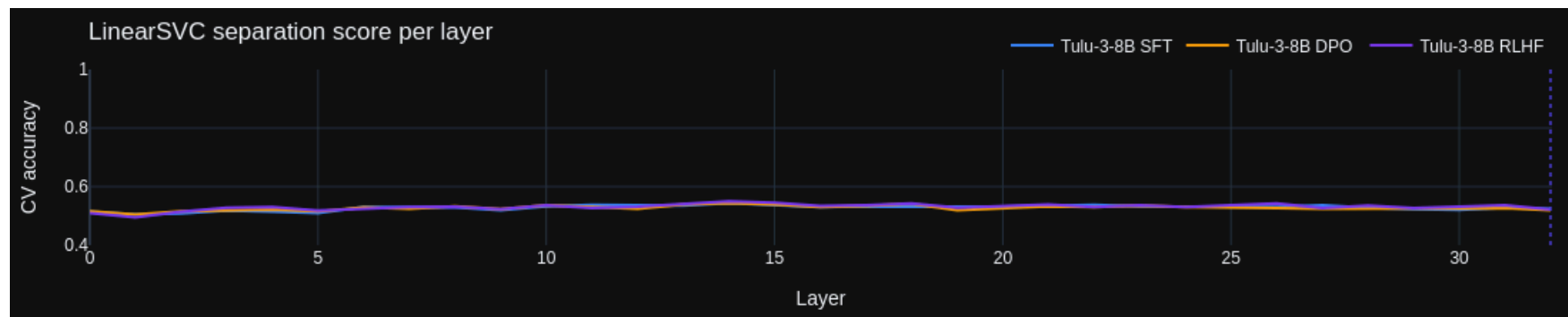
Part 2 — Anisotropy



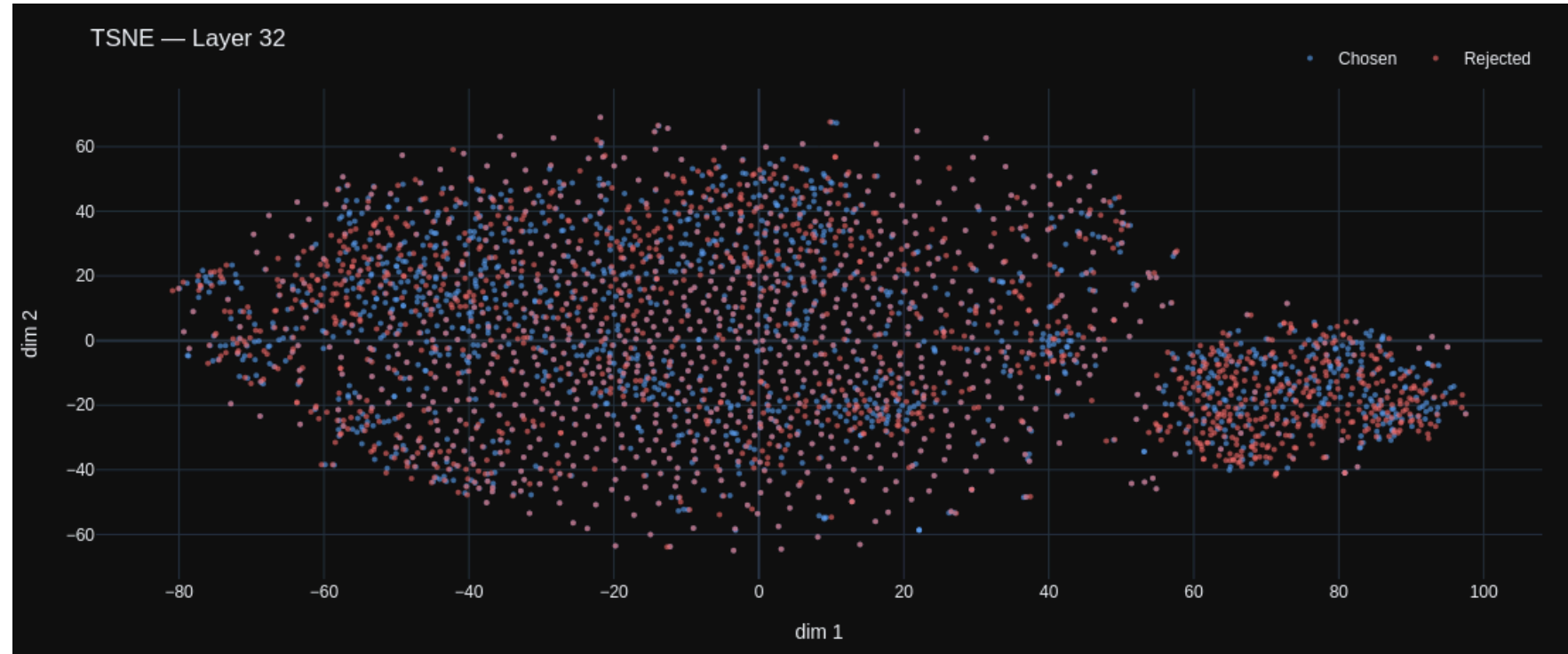
Anisotropy — dips mid-stack, rises again late



Effective rank — dimensionality usage per layer



LinearSVC accuracy — pinned near chance (0.5)



t-SNE, layer 32 — chosen/rejected heavily overlap

Insight: RLHF does not carve a linear “preference direction” We asked: does alignment training create a direction in the residual stream that separates chosen from rejected? LinearSVC and Cohen’s d stay flat and near-chance at **every** layer, in **all three** checkpoints — SFT, DPO, RLHF.

Insight: But the representation space is not static Linear CKA shows real structural reorganization — concentrated almost entirely at **SFT → DPO** (CKA drops to 0.976 by layer 32). The **DPO → RLHF** step barely moves the geometry further (CKA stays above 0.999) — most of the “alignment work” already happened before PPO.

Remark Anisotropy and effective rank both shift in that same SFT→DPO window — consistent with one genuine representational transition, not three separate ones. It just isn’t a transition any single linear probe can read.

*Alignment reshapes geometry **globally**, not by writing a single readable “preference axis” — evidence for treating RLHF’s effect as a geometric audit, not a steering vector to extract.*

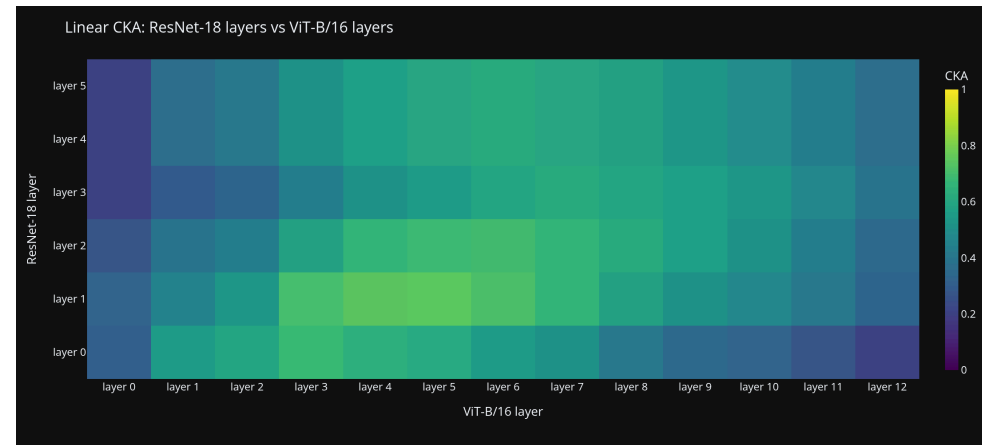
Setup · DINOv2 + Llama · MS-COCO · multi-class prediction

Both models trained on **same task** (object category prediction) · measure CKA · fine-tune with loss term encouraging similarity to the other model's representation · compare

CKA high before fine-tuning: models converge inherently (**Platonic Representation Hypothesis**)

CKA rises after fine-tuning: cross-modal signal is genuinely new information

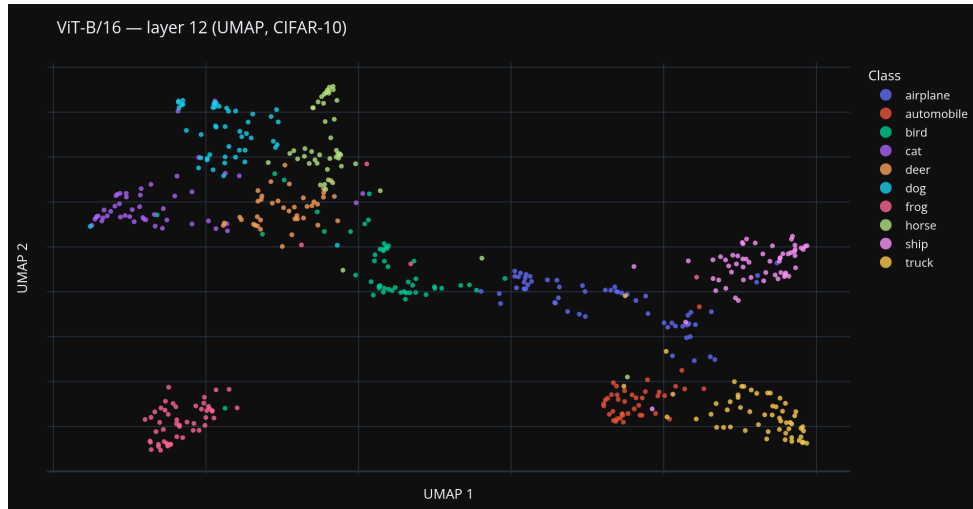
Layer-wise CKA map shows **where** shared meaning lives · UMAP + dashboard to visualize



Interactive Plotly Dash · CKA heatmap

Next steps — toward final

- Cross-modal CKA: DINOv2 vs. Llama on MS-COCO pairs
- **CLIP (ViT-B/32)** as upper bound — explicitly trained cross-modal alignment
- Compare base vs. Instruct CKA scores



Interactive Plotly Dash · CKA heatmap · UMAP layer slider · LinearSVC score

Project deliverable

The dashboard **is** the deliverable — interactive visualization of all findings: layer slider, CKA heatmaps, UMAP scatter, SVC score

Perspective

Extend to RLHF training **snapshots** — plot cross-modal CKA vs. training step to see if alignment shifts **during** RLHF

He, Trott, Khosla (2025)

Shared Latent Representations across Vision and Language

arXiv:2509.20751 · **Anchor paper**

Kornblith et al. (2019)

Similarity of Neural Network Representations Revisited

ICML · arXiv:1905.00414 · **CKA method**

Christiano et al. (2017)

Deep RL from Human Preferences

NeurIPS · arXiv:1706.03741 · **RLHF foundations**

Lambert et al. (2024)

Tulu 3: Pushing Frontiers in Open Language Model Post-Training

arXiv:2411.15124 · **Models used for Part 2**

Kucukahmetler et al. (2026)

Relative Geometry of Neural Forecasters

TMLR · arXiv:2602.15676

Ouyang et al. (2022)

Training LMs to Follow Instructions with Human Feedback

NeurIPS · arXiv:2203.02155 · **InstructGPT / RLHF**

McInnes et al. (2018)

UMAP: Uniform Manifold Approximation and Projection

arXiv:1802.03426 · **Dim. reduction method**